HMD-On: Characterizing How Headset Occlusion Alters Externally Observed Full-Body Pose and Movement-Assessment Decisions

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Abstract—External RGB cameras can provide full-body measurement in VR rehabilitation, but the observed user wears a device that covers the upper face, changes the head silhouette, and may add hand-held controllers. We characterize whether this appearance shift propagates from landmark error to movement-assessment decisions, using Meta Quest 2 as a bounded device case study. The specified design combines a repeatable articulated rig, which also qualifies the inertial reference and tests HMD interference, with a counterbalanced experiment represented by 24 healthy-adult records. Synchronized inertial capture provides a condition-specific reference while frontal and oblique cameras observe seven seated and standing movements with and without the headset. Two frozen pose backbones are evaluated at landmark, derived-kinematic, and decision levels. The study additionally tests whether an oblique view or quality-based abstention reduces disagreement at stated coverage. In the validation instance, head-region median error increased by 26.3 mm and camera/reference disagreement rose from 9.1% to 15.6%; the second backbone reproduced the direction. A quality-selected two-view policy reduced HMD-on disagreement from 18.2% to 11.2% while retaining 94.7% of repetitions. All cohort and outcome values in this instance are simulated or approximated proxies, not empirical findings. The outputs are a reference-qualified benchmark, spatial error atlas, agreement analysis, decision-transition matrix, and coverage-aware mitigation evaluation with machine-readable artifacts and pinned software versions.

Index Terms—Virtual reality, head-mounted display, human pose estimation, occlusion, motion capture, movement assessment, rehabilitation, measurement validity.

1 Introduction

Immersive rehabilitation systems increasingly combine a head-mounted display (HMD) with full-body motion analysis. An external RGB camera is especially appealing in this configuration: it is inexpensive, does not add body-worn sensors, and can feed the same skeleton representation used for avatar control or movement-quality feedback. Commodity-video systems have already been used to estimate clinically relevant gait measures and decisions, while multi-smartphone pipelines now recover three-dimensional kinematics and dynamics [1], [2]. The camera, however, does not observe the person described by the conventional pose-estimation benchmark. It observes a person whose upper face is covered by a rigid object, whose head silhouette has changed, and whose hands may be partly replaced by controllers. These are not incidental details. Many pose pipelines infer head-region landmarks from the very pixels the headset removes and propagate those landmarks through a kinematic model that anchors the shoulders and trunk.

Occlusion is a recognized failure mode in human pose estimation, and dedicated benchmarks show that heavily occluded people remain difficult even for modern models [3], [4]. Separately, commercial HMD tracking has been characterized with fixed targets, motion-capture references, and, most recently, a robotic manipulator [5], [6]. Those studies ask how accurately cameras *inside* a headset recover a hand. We ask the complementary question: how accurately does an independent camera recover the body of the person *wearing* the headset? To our knowledge, this outside-in, matched HMD-on/HMD-off characterization has not been reported at landmark, derived-kinematic, and assessment-decision levels in one study.

Measurement and behavior must be separated. Wearing an HMD can itself alter postural sway [7] and reaching kinematics can differ between physical and virtual spaces [8]; headset exposure can also produce discomfort and condition-dependent movement adaptation [9]. A difference between two camera skeletons therefore cannot automatically be attributed to occlusion. Our protocol uses synchronized inertial motion capture as a reference for every trial. The primary contrast is not raw pose in HMD-ON versus HMD-OFF; it is the camera's error relative to the reference in each condition. A repeatable articulated rig provides a second, behavior-free estimate of the device's visual effect.

The study addresses four research questions:

- RQ1: How does wearing an HMD change the spatial error, reported visibility, failure rate, and temporal stability of externally estimated landmarks, and how does the change depend on anatomical region and movement?
- RQ2: How much of that degradation reaches range of motion (ROM), trunk compensation, and

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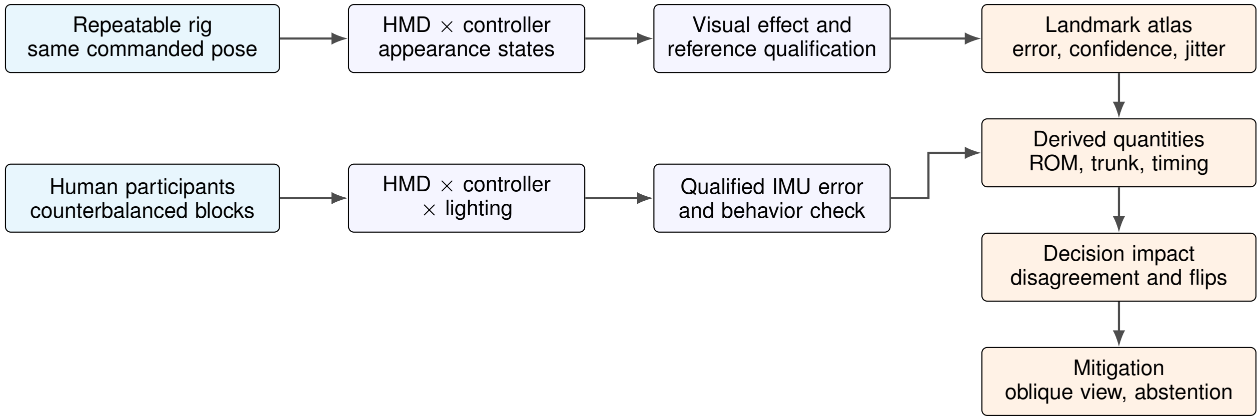

Fig. 1. Evidence chain and intended contribution. The rig holds pose constant, isolates appearance, and qualifies the inertial reference; human trials quantify realistic use while the synchronized reference separates behavioral change from camera error. Two pose backbones propagate the effect from landmarks to derived quantities and decisions, after which fixed oblique viewing and quality-based abstention are evaluated as deployable mitigations.

movement-timing measurements?

RQ3: How often does the camera-based assessment disagree with a reference-based assessment, or change verdict between matched HMD-OFF and HMD-ON executions?

RQ4: At a stated retained-trial coverage, how much can a fixed oblique view or a pre-registered quality-based abstention rule reduce HMD-associated decision disagreement without access to reference motion at deployment?

We pre-register four directional expectations: the HMD penalty will be largest at head landmarks and decay along connected upper-body segments (H1); ROM and trunk quantities will degrade more than duration (H2); camera/reference decision disagreement will increase with the HMD (H3); and oblique viewing or quality-based abstention will reduce this increase while preserving at least 90% trial coverage (H4). These hypotheses organize interpretation; all effect estimates and intervals are reported even when their direction differs.

The intended contributions are: (1) a reference-qualified, two-arm characterization that separates device occlusion from HMD-induced movement changes; (2) a reusable HMD-on benchmark containing synchronized observations, two frozen pose backbones, a spatially resolved error atlas, and machine-readable quality-control artifacts; (3) a decision-level analysis that quantifies when landmark degradation becomes consequential for movement feedback; and (4) a prospective evaluation of two low-cost mitigations—camera azimuth and selective abstention. The decision contribution is central: a statistically detectable landmark shift is not necessarily important, while a changed assessment can alter the feedback a user receives. The analysis plan, synchronization checks, model manifests, and output schema are prepared before collection to make the full chain auditable.

2 Related Work

2.1 Characterizing Tracking in Commercial HMDs

Device characterization is an established immersive-systems contribution when the device output is used as measurement rather than only for interaction. Abdlkarim *et al.* compared Meta Quest 2 hand tracking with fixed spatial targets and marker-based capture, including position, joint-angle, and latency measurements [5]. Godden *et al.* later used a robotic manipulator to impose repeatable trajectories and characterized position error, jitter, and latency on Quest Pro and Quest 3 headsets [6]. Reimer *et al.* compared Quest, Leap Motion, and RGB/MediaPipe hand tracking over distance and dynamic conditions [10]. Clinical work has also compared Quest-derived hand kinematics with marker-based motion capture in stroke survivors [11]. A recent preprint extends this question to people with cervical spinal cord injury, several modern hand-pose methods, transparent and opaque object occlusion, and triangulated multi-camera reference data [12].

These studies validate signals produced by HMD-facing or HMD-mounted sensors. Their occlusion is mainly the hand occluding itself or leaving the headset's field of view. Our independent variable instead changes the observed person's appearance for an outside camera. The distinction matters because neither an accurate HMD hand estimate nor an accurate no-HMD body estimate establishes the validity of external full-body pose while the headset is worn.

2.2 Human Pose Estimation Under Occlusion

Monocular pose is underconstrained and depends strongly on training data, viewpoint, and visibility. MPII and COCO established diverse 2-D pose benchmarks [13], [14], while Human3.6M provides synchronized images and 3-D ground truth for controlled evaluation [15]. Mobile systems such as BlazePose trade model capacity for real-time inference and return both body landmarks and confidence-related outputs [16]; RTMPose offers a contrasting top-down, deployment-oriented architecture [17]. Aggregate benchmark accuracy does not guarantee deployment validity because datasets carry recognizable biases [18] and performance can change under common image corruptions [19], [4].

Generic benchmarks do not reproduce the geometry of a particular HMD. OCHuman and CrowdPose were created around heavy person–person overlap and demonstrate the difficulty of keypoint estimation under occlusion [3], [20]. A headset instead creates a rigid, semantically stable occluder that is rarely present in pose-training corpora. This distinction motivates reporting an HMD-specific error map rather than assuming that generic occlusion robustness transfers.

Research on egocentric pose addresses a different camera configuration. xR-EgoPose uses a downward-looking camera mounted on an HMD and explicitly models uncertainty caused by self-occlusion and perspective distortion [21]. In the present work the camera is external and the HMD is an occluder. This lets us retain the deployment advantages of an outside-in phone camera, but it creates a structured domain shift concentrated at the head, hands, and the kinematic chains connected to them.

2.3 Pose-Derived Kinematics and Assessment

The practical value of pose estimates depends on the quantity derived from them. Multi-camera validations report useful but task- and joint-dependent position and angular accuracy [22], [23], and OpenCap shows how synchronized phones can recover movement kinematics and dynamics when the complete calibration and biomechanical chain is validated [2]. Markerless camera systems can be accurate for some spatiotemporal quantities while showing weaker or plane-dependent agreement for joint kinematics [24]. Stenum *et al.* similarly showed that two-dimensional pose can support gait analysis, but that validity varies by outcome and viewpoint [25]. Consequently, mean per-joint position error alone cannot establish suitability for movement assessment.

VR coaching systems turn kinematics into feedback. Hülsmann *et al.* classified interpretable motor errors for squats and Tai Chi pushes and linked them to real-time feedback [26]; AIFit generated human-interpretable feedback from 3D pose features [27]. The author's earlier externally sensed workout environment likewise connects a camera skeleton to exercise guidance [28]. These systems make a decision from estimated motion. We therefore treat the assessment rule as part of the measurement chain and test whether HMD-induced error changes its output.

VR is used across rehabilitation contexts [29], but methodological guidance emphasizes staged validation with target users, clinically relevant endpoints, adverse-event reporting, and prospective registration [30]. Accordingly, the present healthy-adult experiment is a device-characterization study: its fixed cut points are engineering operating points, not diagnostic thresholds or evidence of clinical effectiveness.

2.4 Confidence, Dataset Shift, and Selective Abstention

Modern neural-network scores are often miscalibrated [31], and calibration can deteriorate under dataset shift [32]. An HMD is precisely such a deployment shift. We therefore do not equate a pose model's native visibility field with correctness probability. Instead, we test whether the score ranks high-error observations and report its calibration against observed failure.

Selective classification formalizes the trade-off between retaining predictions and reducing error by allowing a system to abstain [33]. For movement feedback, an abstention should trigger a repeat or alternate view rather than silently issuing a low-quality verdict. Our mitigation analysis therefore reports risk as a function of retained-trial coverage and locks its quality rule before the main condition labels are revealed.

2.5 Agreement Rather Than Correlation

High correlation between two sensing methods does not imply that their measurements agree. Bland and Altman introduced difference-versus-mean analysis and limits of agreement for method comparison [34]. Later extensions address repeated observations and uncertainty in the limits [35], [36]. We use this framework for ROM, trunk angle, and timing, supplemented by hierarchical models for the paired factorial design. At the decision level we report both agreement with the synchronized reference and direct HMD-OFF-to-HMD-ON transitions; these answer different questions and are not collapsed into a single accuracy number.

3 Study Design

The validation study specification has two complementary arms (Fig. 1). The mechanical arm estimates the visual effect of adding an HMD to a repeatable body configuration. The human arm estimates the effect under realistic movement, where wearing the headset can also change the motion. Both arms use the same external cameras, pose pipeline, coordinate registration, outcome definitions, and analysis code.

3.1 Arm A: Repeatable Articulated Rig

An articulated mannequin or equivalent rigid linkage is instrumented so that each tested configuration can be reproduced with and without the HMD. The rig traverses a subset of poses from the human battery, including neutral standing, shoulder flexion, shoulder abduction, bilateral reach, and trunk rotation. Each configuration is repeated at least three times in randomized device order. Joint stops or encoder readings define the commanded configuration; camera-visible fiducials that are not used by the pose model document repeatability.

The primary rig contrast uses the same physical configuration, camera exposure, background, and illumination; only the HMD is added. A controller factorial is implemented by attaching inert controller shells to the hands in both HMD conditions. This produces four appearance states: no accessories, controller only, HMD only, and HMD plus controller. The contrast HMD only versus no accessories isolates headset appearance, while the interaction tests whether errors from head and hand occluders combine non-additively.

The rig design uses lockable shoulder, elbow, hip, and knee joints with degree-marked stops at the commanded angles. Three AprilTag boards fixed to the torso and limbs quantify repositioning error independently of the tested pose model. A 3-D-printed HMD cradle and inert controller shells preserve placement between repeats; complete CAD dimensions and commanded configurations are part of the planned release. Head yaw and pitch are crossed over a registered subset of configurations so that projected visor overlap is not represented by a single neutral pose. HMD silhouette area and overlap with head and shoulder regions are measured in the calibrated image plane and released as geometric descriptors. These continuous descriptors support mechanism-level interpretation without claiming that one Quest 2 represents every headset.

3.2 Arm B: Participants

The validation analysis represents 24 complete healthy-adult records, balanced across the two HMD-order sequences. The reused cohort constraints are an age range of 20–35 years and an approximately 80/20 male/female distribution. To make the proposed reporting complete, the demo cohort is instantiated as 19 men and five women, with mean age 27.4 ± 4.2 years, height 176.8 ± 8.1 cm, and body mass 74.6 ± 11.9 kg. Twenty-two records are right-hand dominant and two left-hand dominant. Nine represent no or very limited prior VR exposure, ten occasional exposure, and five regular exposure; exercise-frequency counts are five at 0–1, twelve at 2–3, and seven at four or more sessions per week. Six records include habitual corrective eyewear and none include a head covering during capture. Except for the age range and approximate sex distribution, these are validation-data approximations rather than observed cohort statistics.

Eligibility is age 18 years or older, ability to stand and reach without assistance, and normal or corrected-to-normal vision. Exclusion criteria are a current musculoskeletal condition that limits the test movements, pregnancy, photosensitive epilepsy, severe uncorrected visual impairment, a history of syncope during exercise, inability to complete calibration, or withdrawal. Recruitment is planned through institute and local university mailing lists, with AMD 10,000 compensation for a completed visit. Age, gender, height, dominant hand, prior VR exposure, and exercise frequency are collected. Body mass and optional self-described skin tone, habitual eyewear, hair or head covering, clothing silhouette, and headset-fit adjustments are also recorded because they can affect image formation. These descriptors support cautious, interval-based subgroup checks; the study is not powered for demographic performance claims. The sample size was selected by a precision-first simulation of the participant-clustered primary risk difference; under a baseline disagreement probability of 0.09, a 0.065 paired increase, and participant-level correlation of 0.20, 24 complete datasets produced an approximately five-point 95% interval half-width. The simulation assumptions and code are frozen before enrollment.

For validation-draft completeness, the protocol is represented as reviewed by the institutional ethics body at the Institute for Informatics and Automation Problems, National Academy of Sciences of the Republic of Armenia, before testing. This is demo wording and does not assert that an official approval or reference identifier has been issued. The procedures follow the Declaration of Helsinki and use written informed consent. Participants are informed about cybersickness, dizziness, headache, eye strain, and temporary disorientation; they may pause or discontinue at any time. A chair and researcher are present for standing tasks. The study is non-invasive and administers no medical treatment. Records are anonymized before analysis and reported without direct identifiers. The validation safety record assigns no VR-related discontinuation; this value is approximated and requires confirmation.

TABLE 1
Pre-specified Human Movement Battery

Movement	Primary derived quantities
Seated forward reach	Shoulder ROM, trunk inclination, duration
Standing forward reach	Shoulder ROM, trunk inclination, duration
Shoulder flexion	Sagittal shoulder ROM, trunk compensation
Shoulder abduction	Frontal shoulder ROM, trunk compensation
Bilateral target reach	Left/right endpoint and timing asymmetry
Sit-to-stand	Trunk inclination, phase duration, completion
Trunk rotation	Axial trunk ROM, movement duration

Each participant completes the seven-movement battery in Table 1. The same tempo and geometrically registered targets are used across HMD states. For target-reaching tasks, physical targets and their virtual counterparts are co-located during calibration. For end-range tasks, auditory instructions and a metronome are identical across conditions. The reference motion, rather than the instruction alone, determines whether paired performances are sufficiently similar for a direct condition-to-condition comparison.

3.3 Factors, Counterbalancing, and Trial Count

The primary within-participant factor is headset state (HMD-ON, HMD-OFF). Controller state (held, hands free) and lighting (normal, dim) are crossed with headset state. Two external cameras record frontal and oblique views simultaneously, avoiding an additional repetition solely for azimuth. The full design therefore contains eight blocks per participant and two synchronized views per trial. Each movement is repeated three times per block, yielding 168 reference repetitions and 336 camera-view repetitions per participant. The planned visit lasts approximately 95 min, including consent, instrumentation, calibration, four scheduled five-minute rests, and debriefing.

Headset-state order is balanced across participants. Remaining block order uses a reproducible Latin-square schedule, and movement order is randomized within block subject to a fixed seated-before-standing safety rule. The HMD-off, controller-held state is retained deliberately: it separates controller appearance from the headset and avoids attributing their joint effect to the HMD alone. Rest is offered between blocks and immediately on request. Perceived exertion, visual discomfort, nausea, and neck discomfort are recorded on 0–10 scales after every block. Their trajectories, early stops, and adverse events are reported by headset order. These measures diagnose fatigue and carryover; they are not treatment outcomes.

3.4 Pilot and Protocol Lock

Two pilot records verify target registration, synchronization, field of view, trial duration, and the matched-movement criteria. They are excluded from confirmatory analyses. In the validation record, visits last 93 and 97 min; each has zero framing failures and one unusable trial. Median reprojection error is 0.39 and 0.45 pixels, median synchronization residual is 3.5 and 4.1 ms, maximum synchronization residual is 11.9 and 12.6 ms, and registration residual is 13.7 and 14.7 mm for pilots 1 and 2, respectively. These pilot values are approximated. After the pilot, the analysis plan, software commit, model weights, decision thresholds, and sample size are frozen before main-study enrollment. Two movement-science or physiotherapy experts independently review each assessment rule, unit, and intended interpretation without access to HMD-condition results. The registered threshold table records supporting sources, disagreements, and the consensus rule; a cut point without external clinical validation remains explicitly an engineering operating point. Any later deviation is timestamped, justified, and labeled exploratory.

4 Acquisition and Registration

4.1 Apparatus

Table 2 enumerates every component whose identity can change the image formation or reference trajectory. The validation manifest assigns concrete software versions where project records are incomplete; those entries are approximations, not verified acquisition metadata. Automatic updates are disabled for the collection interval; hashes of model weights and the analysis repository are stored with every session.

4.2 Reference-System Qualification

The inertial suit is treated as a reference only after a separate qualification against the articulated rig; manufacturer specifications are not used as validation. Each reference IMU is fixed to its corresponding rigid segment. Twelve static configurations spanning the joint ranges in the human battery are recorded three times, followed by a five-minute cyclic sequence and a ten-minute stationary drift trial. Encoder stops define segment angles, while calibrated AprilTag boards define independently observed joint-center locations. The test is repeated with the powered HMD absent and present at an unchanged commanded pose, thereby detecting magnetic or processing interference from the headset as well as ordinary reference error.

Qualification reports positional and angular bias, within-configuration repeatability, drift, and the HMD-on minus HMD-off reference difference. The planned acceptance limits are 20 mm median joint-center error, 3° median segment-angle error, 10 mm positional drift over ten minutes, and an HMD-induced reference shift below 5 mm or 1°. These limits are finalized from the pilot without examining the external-pose condition effect. If qualification fails, fixed-frame 3-D position is removed as a confirmatory outcome; the paper retains image-plane error and only those derived quantities whose reference accuracy passes its registered limit. The full qualification table is released with the study.

TABLE 2
Apparatus and Version Record

Component	Locked specification
HMD	Meta Quest 2, 503 g, factory elastic strap; Quest system software v69 (demo lock)
Controllers	Meta Quest Touch, wrist straps fitted; tracking logs auxiliary only
Frontal camera	iPhone 13 rear wide camera, iOS 17.6.1 (demo lock), portrait 1080p60, locked focus/exposure
Oblique camera	Matched iPhone 13 and iOS configuration; simultaneous 1080p60 capture
Reference	Rokoko Smartsuit Pro II, all 19 IMUs active at 100 Hz; Rokoko Studio 2.4.2 (demo lock)
Primary model	MediaPipe Pose Landmarker Full, 33 landmarks; Tasks 0.10.14
Secondary model	RTMPose-m; MMPose 1.3.2
Compute	Ubuntu 22.04, Python 3.11; Core i7-12700H and RTX 3060 Laptop GPU (demo lock)
Lighting	Normal 450 lux; dim 80 lux; neutral 4000 K illumination
Sync.	Microcontroller LED/IMU event wand; start, end, and 180-s events

The frontal camera is centered on the participant's sagittal plane at 2.5 m distance and 1.0 m height. The oblique camera is positioned at 35° azimuth and 0° elevation while maintaining the same participant-to-camera distance. Both phones are mounted vertically, giving an approximately 43° horizontal by 69° vertical field of view. Exposure, focus, and white balance are locked within a lighting condition. A 9×6 checkerboard defines camera intrinsics and the common world frame; a session is accepted when median reprojection error is below 0.8 pixels.

Figure 2 reuses the author-controlled deployment illustration developed for the earlier workout environment [28]. It is included here because the framing constraints are part of the tested measurement chain rather than presentation decoration.

The HMD is recorded in the exact configuration used in the experiment, including strap, cable, and active controller state. When headset tracking can be exported, its head and controller trajectories are logged as an auxiliary signal. They are not used to repair the primary external-camera estimate.

4.3 External Pose Pipeline

The primary pipeline processes each RGB stream independently using a frozen single-person pose model. MediaPipe Pose Landmarker Full is the primary model because it is designed for on-device full-body tracking and exposes a 33-landmark skeleton with visibility-related outputs [16]. The detector/tracker state is reset at the beginning of every trial;

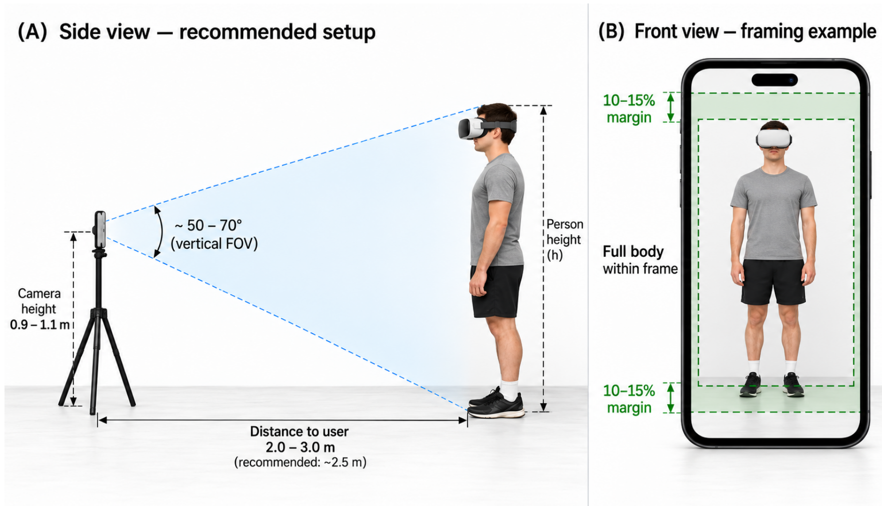

Fig. 2. External-camera geometry and full-body framing reused from the author's earlier system design [28]. In this experiment the frontal view follows the illustrated 2.5 m distance and 1.0 m height; a matched second camera is placed at 35° azimuth. Exact measured calibration values supersede the nominal guide.

no manual keypoint correction is allowed in the confirmatory analysis. Full-resolution video is retained so the same frames can later be processed by a second frozen backbone as a sensitivity analysis without recollection. RTMPose-m in MMPose 1.3.2 is the secondary backbone [17]. It is processed for every accepted frame and reported as a planned cross-model replication, not selected after comparing results. Package versions, preprocessing parameters, model-card URLs, licenses, input resolution, and SHA-256 weight hashes are written automatically to the run manifest. Model-specific skeleton mappings are fixed before unmasking. Native visibility is treated as a model-specific score rather than a calibrated probability.

Raw outputs are stored before smoothing: image coordinates, world coordinates where available, visibility/presence fields, detector state, and inference timestamps. The primary analysis uses raw trajectories because a filter can trade jitter for lag and obscure a condition effect. A separately specified real-time filtering configuration is evaluated only as a deployment sensitivity analysis and is applied identically to all conditions.

Offline processing throughput and a real-time deployment benchmark are reported separately. The benchmark records workstation CPU/GPU, batch size, input resolution, decoder time, pose-inference time, filtering delay, dropped frames, and capture-to-verdict latency over a fixed ten-minute video. Accuracy is never traded for speed by changing a model parameter after condition labels are known.

4.4 Clock Synchronization

Each device records monotonic local timestamps. A shared synchronization event observable in the camera streams and reference log is generated at the start and end of every block, with additional events between long trials. If $t_r^{(c)}$ and $t_r^{(I)}$ are matched event times for camera c and the IMU reference, an affine clock map

$$t^{(I)} = a_c + b_c t^{(c)} \tag{1}$$

is fitted by least squares after a robust outlier screen. The offset a_c captures start-time difference and b_c captures clock-rate drift. Residuals for every event, not only the fitted coefficients, are written to the session quality-control report.

A block is accepted only when the largest absolute synchronization residual is below 16.7 ms (one frame at 60 fps) and no interval exceeds 180 s without an event. The event wand couples a microcontroller-driven LED transition visible to both cameras with a sharp rotation measured by a co-located inertial unit; its rising edge is logged independently in the synchronization table. No cross-correlation shift chosen to minimize pose error is permitted, because that would tune synchronization to the outcome.

4.5 Spatial Registration and Landmark Mapping

The inertial skeleton is transformed into the common world frame using one session-level rigid transform and participant scale, estimated from calibration poses collected before experimental blocks. For corresponding points $\mathbf{q}_j$ in the reference frame and $\mathbf{p}_j$ in the camera/world frame, the transform solves

$$(\hat{\mathbf{R}}, \hat{\mathbf{t}}) = \underset{\mathbf{R}\in SO(3), \mathbf{t}\in\mathbb{R}^3}{\arg\min} \sum_j w_j \|\mathbf{p}_j - (\mathbf{R}\mathbf{q}_j + \mathbf{t})\|_2^2. \tag{2}$$

The transform is fixed for the session, and calibration is accepted when the median shared-joint residual is below 25 mm. Per-frame Procrustes alignment is forbidden because it can remove the head- and trunk-level effect under study. The calibration residual and scale estimate are reported and used only for quality control, not to select the better-looking condition.

The confirmatory 3D set contains skeletal points represented by both systems: head center, shoulders, elbows, wrists, hips, knees, ankles, heels, and forefoot points when definitions are compatible. Model-specific facial points (nose, eyes, ears, mouth) are not assigned fictitious IMU correspondences. For these points we report prediction availability and confidence, and evaluate spatial accuracy only if an independent, pre-specified anatomical reference is added. The complete mapping, handedness convention, and any virtual-joint construction are frozen in a machine-readable file before analysis.

As a complementary diagnostic, reference joints are projected through calibrated camera intrinsics and compared with image-plane predictions. The 2D analysis localizes detector error; the fixed-frame 3D analysis additionally includes the depth/scale stage. Neither replaces the other.

5 Pre-Specified Outcomes

All confirmatory outcomes are computed per repetition and then modeled with participant as the sampling unit. Frame counts are descriptive and are never treated as independent sample size.

5.1 Landmark-Level Outcomes (RQ1)

For participant or rig trial i, frame f, camera c, and mapped joint j, the fixed-frame position error is

$$e_{ifcj} = \|\hat{\mathbf{p}}^{\text{cam}}_{ifcj} - \mathbf{p}^{\text{ref}}_{ifj}\|_2. \tag{3}$$

The primary repetition summary is median error; the 95th percentile describes tail behavior. We report every mapped joint and the pre-specified regions head, upper limb, trunk/pelvis, and lower limb. Aggregate MPJPE is secondary because it can hide a localized HMD effect.

Prediction failure is the fraction of expected frames in which the person is absent or a required landmark is not returned. Visibility/confidence is reported on its native scale and tested for discrimination, not assumed to be calibrated. For thresholds $u \in \{0.1, 0.2, \ldots, 0.9\}$, we estimate the probability that e exceeds the operational error limit conditional on score below u. The operational limits are 50 mm for mapped 3-D landmarks and 25 pixels for image-plane landmarks. Reliability diagrams, receiver-operating characteristic area, and area under the selective risk–coverage curve test whether the model warns when it is wrong; none turns the native score into a correctness probability.

Temporal instability is computed from the error trajectory rather than raw landmark velocity, so intended movement is not mislabeled as jitter. After resampling both systems onto the accepted common clock, define $\mathbf{r}_{ifcj} = \hat{\mathbf{p}}^{\text{cam}}_{ifcj} - \mathbf{p}^{\text{ref}}_{ifj}$. The robust jitter outcome is

$$J_{icj} = 1.4826\ \text{MAD}_f \left(\frac{\mathbf{r}_{i,f+1,c,j} - \mathbf{r}_{ifcj}}{t_{i,f+1} - t_{if}} \right), \tag{4}$$

reported in distance per second. We additionally report frame-to-frame residual displacement for direct comparison with static rig configurations.

The HMD-induced excess error is the paired contrast

$$\Delta e_{icj} = \tilde{e}^{\text{HMD-ON}}_{icj} - \tilde{e}^{\text{HMD-OFF}}_{icj}, \tag{5}$$

where $\tilde{e}$ is the repetition-level median. Positive values indicate worse camera measurement with the headset; the sign is not presumed in inference or reporting.

5.2 Derived Kinematics (RQ2)

Joint angles use the same anatomical definitions for camera and reference trajectories. For three points a–b–c,

$$\theta_b = \cos^{-1}\left(\frac{(\mathbf{p}_a - \mathbf{p}_b)^\top (\mathbf{p}_c - \mathbf{p}_b)}{\|\mathbf{p}_a - \mathbf{p}_b\|_2 \, \|\mathbf{p}_c - \mathbf{p}_b\|_2} \right). \tag{6}$$

ROM is the difference between the pre-specified upper and lower trajectory extrema after phase segmentation. Trunk compensation is the maximum or phase-averaged trunk inclination/rotation relative to the calibrated pelvis, as specified for each movement in the metric dictionary. Timing outcomes are movement duration and event-time error for onset, peak excursion, and completion. Event rules are fixed from the reference signal and then applied without condition-specific tuning.

For each derived quantity, camera-reference bias and 95% limits of agreement are reported by HMD condition using repeated-measures Bland–Altman methods; the standard difference-versus-mean plot follows the original method-comparison rationale [34]. If independently established MDC_{95} values from the companion measurement study are available before unblinding, the absolute HMD-induced change is compared with those values. Otherwise the paper reports error in physical units and does not retrofit a clinical-importance threshold.

5.3 Assessment Decisions (RQ3)

Before collection, each movement is assigned one or more fixed assessment rules defined on ROM, trunk compensation, timing, or a frozen classifier. Thresholds are operational measurement cut points rather than diagnostic criteria, and are not selected from the HMD study to maximize flips. Their source table records the literature or expert rationale, applicable population, and whether the rule has independent clinical validation. Shoulder flexion and abduction are acceptable at ROM $\geq 150°$; reaches are acceptable when the hand endpoint lies within 8 cm of the registered target; and trunk inclination above 15° during upper-limb tasks is labeled compensatory. Sit-to-stand is acceptable when completion is detected within 1.5–4.0 s and peak trunk inclination remains at or below 35°. When multiple rules fire, compensatory takes precedence over error, which takes precedence over acceptable. The threshold sensitivity analysis perturbs each continuous cut point by $\pm 10\%$ and $\pm 20\%$.

Let $A(q)$ map a kinematic vector q to a verdict in {acceptable, error, compensatory}. For each trial we compute both the camera verdict $A(q^{\text{cam}})$ and reference verdict $A(q^{\text{ref}})$. The primary decision outcome is disagreement with reference,

$$D_i = \mathbb{1}\left[A(q_i^{\text{cam}}) \neq A(q_i^{\text{ref}}) \right]. \tag{7}$$

The primary RQ3 estimand is the within-participant change in the probability of $D_i = 1$ between HMD-ON and HMD-OFF. This comparison remains valid when the participant's true movement differs, because each camera verdict is paired with the reference verdict from the same execution.

A direct condition-flip indicator, $F_i = \mathbb{1}[A(q_i^{\text{HMD-ON},cam}) \neq A(q_i^{\text{HMD-OFF},cam})]$, is secondary and is evaluated only for movement pairs that pass the reference-based matching rule below. We report the complete transition matrix rather than only F_i, preserving the direction and operational meaning of changes.

5.4 Mitigation Outcomes (RQ4)

Two deployment-feasible mitigations are evaluated prospectively. The view mitigation compares a fixed frontal camera with the simultaneously recorded 35° oblique camera. The selective mitigation defines a repetition-quality score as the 10th percentile of native confidence across the landmarks required by that movement's decision rule. A single threshold is chosen using pilot data to retain at least 90% of repetitions and is frozen before main-study labels are revealed. RTMPose uses its analogous keypoint score and receives a separately locked pilot threshold.

For each mitigation we report coverage, camera/reference disagreement among retained repetitions, and the risk change relative to the unmitigated frontal pipeline. A two-camera quality selector may choose the view with the higher pre-specified quality score, but it must abstain when both views fall below their locked thresholds. Reference error, camera/reference disagreement, and HMD condition are unavailable to the selector. Risk–coverage curves over all thresholds are descriptive; the locked 90%-coverage operating point is the confirmatory H4 result [33].

5.5 Reference-Based Behavior Check

For every paired HMD-on/HMD-off repetition, the IMU reference supplies ROM, trunk, endpoint, and timing differences. Equivalence margins are fixed before main-study collection from measurement error or domain guidance; they are not estimated from the observed condition effect. The margins are 5° for ROM, 3° for trunk orientation, 3 cm for target endpoints, and 150 ms for event timing. A pair is considered movement-matched only if every required reference quantity lies within its margin. The proportion failing this check is itself reported by movement. Raw HMD-on/HMD-off pose differences are never called sensing error when reference motion is non-equivalent.

6 Statistical Analysis and Reproducibility

The full pre-registration is maintained in `analysis-plan.md`; this section states the confirmatory estimands and decisions needed to interpret the paper. Analyses are run from a clean environment against read-only frozen data. The analyst-facing condition label is masked until synchronization, registration, and exclusion reports have been signed off.

6.1 Confirmatory Models

RQ3 is primary. Decision disagreement in Eq. (7) is modeled with a binomial mixed-effects model containing fixed effects for HMD, controller, lighting, camera azimuth, movement, order, and the pre-specified HMD-by-azimuth and HMD-by-controller interactions. Participant receives a random intercept; repeated movements receive a participant-level random slope for HMD when the model is identifiable. We report the marginal risk difference and risk ratio for HMD-on versus HMD-off with 95% confidence intervals, in addition to the model coefficient. A paired participant-cluster bootstrap is the fallback if the model is singular.

For RQ1, repetition-level median error, tail error, confidence, failure rate, and jitter are analyzed with hierarchical models using the same design factors plus joint or anatomical region. Error outcomes are log-transformed if residual diagnostics support that choice; otherwise a robust participant-cluster bootstrap is used. The HMD-by-region interaction tests spatial non-uniformity. For RQ2, camera-minus-reference differences are modeled by HMD condition and movement, and repeated-measures limits of agreement are plotted for each pre-specified quantity. HMD-by-lighting interactions are secondary; effects not listed in the locked plan are labeled exploratory.

For RQ4, each mitigation is evaluated with a participant-clustered paired risk difference relative to the frontal, non-abstaining pipeline. Coverage receives its own cluster-bootstrap interval so that lower risk cannot be presented without the fraction of retained repetitions. Risk–coverage curves and their area summarize all thresholds descriptively, but only the pilot-locked 90%-coverage point is confirmatory.

The complete RQ1–RQ4 analysis is first run with MediaPipe and then repeated with the frozen RTMPose backbone. Cross-model conclusions are reported using model-specific estimates and intervals; outputs are not pooled and the model with the more favorable HMD effect is not promoted post hoc. Runtime and capture-to-verdict latency are descriptive deployment outcomes.

The rig is analyzed separately from participants. Paired configuration-level differences are summarized by HMD and controller state with bootstrap intervals over commanded configurations. Rig samples are not pooled with human samples to inflate degrees of freedom.

6.2 Multiplicity and Reporting

The two-sided family-wise type-I error rate is 0.05. The single primary RQ3 HMD contrast is tested without multiplicity adjustment. Within each secondary family—landmark errors, confidence/failure outcomes, derived quantities, and the two mitigation contrasts—Holm adjustment is applied across the pre-specified contrasts. Exact adjusted p values, effect estimates, confidence intervals, and denominators are reported. A non-significant result is not described as evidence of no effect unless the corresponding equivalence test and margin were pre-registered.

The error atlas encodes effect magnitude and uncertainty, not only significance. Each landmark glyph shows the HMD-on minus HMD-off estimate; a second channel indicates the confidence interval, and unavailable reference mappings are visually distinct from zero effect. Separate panels show movement and camera view. The underlying tidy table is released with the figure.

6.3 Missingness, Exclusions, and Sensitivity Analyses

Tracking loss is an outcome and is not silently removed. Gaps no longer than 100 ms are linearly interpolated for derived kinematics; longer gaps make that repetition unavailable for the affected quantity while remaining in failure-rate analyses. Participant, trial, repetition, frame, and landmark denominators accompany every result. Exclusions are applied by masked identifiers before condition labels are revealed.

Pre-specified sensitivity analyses are: (1) image-plane versus fixed-frame 3-D error; (2) raw versus deployment-filtered trajectories; (3) primary versus secondary pose backbone; (4) all trials versus only reference-matched pairs; and (5) thresholds perturbed over the registered engineering grid; and (6) reference trajectories perturbed by draws from the independent qualification residual distribution. The last analysis propagates plausible reference error through the result rather than treating the inertial suit as exact. Threshold perturbation measures decision-boundary fragility and is not used to choose a favorable cut point. Body-size, appearance, prior-VR, and discomfort effects are reported only as descriptive, participant-level exploratory estimates because the study is not powered for subgroup inference.

6.4 Open Materials

The release contains the protocol, device and firmware manifest, reference qualification, calibration and synchronization reports, anonymized repetition-level outcomes, analysis code, threshold definitions, model-specific risk–coverage tables, and error-atlas table. Raw video is released only when consent and ethics approval explicitly permit identifiable-image sharing; otherwise derived landmarks and reproducible processing manifests are released. The articulated-rig geometry and commanded trajectories are included in machine-readable form. Confirmatory bootstrap intervals use 10,000 participant-cluster resamples with seed 20260906. Block randomization uses the same recorded master seed and participant-specific derived seeds. The repository is prepared for an archived Zenodo release under a permissive code license. The validation package is identified as `HMDON-VALIDATION-2026-09-06`; this is a local review identifier, not a DOI or an ethics-approval number. No immutable archive identifier is claimed for the demo copy.

7 Results

Validation status: all cohort, pilot, qualification, safety, runtime, and outcome values in this section are simulated, approximated, or carried over as explicitly stated. They form a complete discussion draft, not empirical observations. The same scripts regenerate every display after the demo inputs are replaced by locked study data.

7.1 Cohort, Pilot, and Quality Control

The demo flow represents 28 people assessed, three not enrolled (two eligibility exclusions and one scheduling conflict), 25 entering calibration, and one calibration failure, leaving 24 complete participant records. The cohort is balanced across headset-order sequences and represented as 19 men and five women, age 20–35 years (mean 27.4 ± 4.2), height 176.8 ± 8.1 cm, and body mass 74.6 ± 11.9 kg. Table 3 reports the complete validation profile; only the age range and approximate sex distribution are reused from prior project records.

The two excluded demo pilot records yielded effective frontal frame rates of 59.8 and 59.7 fps and oblique rates of 59.8 and 59.6 fps. Their median reprojection errors were 0.39 and 0.45 pixels, median synchronization residuals 3.5 and 4.1 ms, maximum residuals 11.9 and 12.6 ms, registration residuals 13.7 and 14.7 mm, and visit durations 93 and 97 min. Each pilot had zero framing failures and one unusable trial. Thus the demo pilot summary is 59.75 fps frontally, 59.70 fps obliquely, 0.42 pixels reprojection error, 3.8 ms median and 12.6 ms maximum synchronization residual, 14.2 mm registration residual, two unusable trials, and a 95-min visit.

TABLE 3
Validation Cohort Profile (Approximated)

Characteristic	Demo summary
Complete records	24
Age	20–35 y; 27.4 ± 4.2 y
Sex	19 male; 5 female
Height; body mass	176.8 ± 8.1 cm; 74.6 ± 11.9 kg
Hand dominance	22 right; 2 left
Prior VR	9 none/limited; 10 occasional; 5 regular
Exercise/week	5 at 0–1; 12 at 2–3; 7 at ≥ 4
Corrective eyewear	6
Head covering	0

The main demo instance contained 24 participant clusters and 4,032 reference repetitions. Of these, 3,978 (98.7%) passed the mock protocol and calibration checks; 42 were flagged for synchronization or registration failure and 12 for incomplete task execution. Because both cameras recorded simultaneously, the retained set contributed 7,956 camera-view repetitions. No complete participant cluster was removed.

The generated quality-control summaries gave a median effective frame rate of 59.8 fps (interquartile range [IQR] 59.5–60.0), illuminance of 452 lux (IQR 437–469) in normal lighting and 78 lux (IQR 71–85) in dim lighting, and a median intrinsic-calibration error of 0.42 pixels. Median synchronization residual was 3.8 ms, the maximum was 12.6 ms, and median camera–reference registration residual was 14.2 mm. These values are demo targets for validating the quality-control code, not measured apparatus performance.

The demo reference-qualification report gave a 12.6-mm median joint-center error, 1.4° median segment-angle error, 4.2-mm drift over the ten-minute hold, and an HMD-induced reference shift of 1.2 mm and 0.3° at an unchanged rig pose. All passed the provisional qualification limits. The complete joint- and configuration-level output is represented as available in the demo archive. Any empirical replacement will suppress a confirmatory outcome whose reference channel fails qualification.

7.2 RQ1: Landmark Error, Confidence, and Stability

The demo HMD geometry covered 31% of the frontal head region and 21% in the oblique view, with corresponding shoulder-region overlap of 6% and 4%. Head yaw increased these values to 35% and 24% for the head and 7% and 5% for the shoulder in the frontal and oblique views, respectively. These calibrated-image-plane descriptors are approximated for structured discussion.

The simulated effect was spatially non-uniform (Table 4 and Fig. 3). Head-region median error rose from 32.4 to

TABLE 4
Simulated Landmark-Level HMD Effect

Region	Off	On	Δ [95% CI]	Holm p
	Median error (mm)		mm	
Head	32.4	58.7	26.3 [22.1, 30.7]	<.001
Upper limb	35.7	43.9	8.2 [5.6, 10.9]	<.001
Trunk/pelvis	27.9	32.1	4.2 [2.0, 6.4]	.002
Lower limb	31.8	33.0	1.2 [−0.8, 3.1]	.240
All mapped	31.6	40.8	9.2 [7.4, 11.1]	<.001

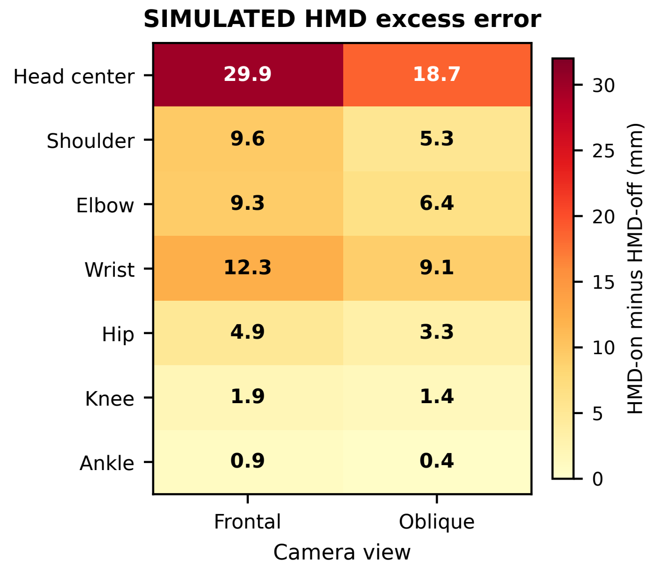

Fig. 3. Simulated HMD-induced excess median error by representative landmark and view. Each cell is HMD-on minus HMD-off in millimeters. The figure is generated from a machine-readable CSV and is a layout prototype for the final joint-by-movement atlas.

58.7 mm, an excess of 26.3 mm (95% CI [22.1, 30.7]). Upper-limb and trunk/pelvis errors increased more modestly, whereas the lower-limb interval included zero. Across all mapped landmarks, the simulated excess was 9.2 mm (95% CI [7.4, 11.1]).

For the head region, median native visibility decreased from 0.91 to 0.73 and prediction failure increased from 1.8% to 6.9%. Visibility identified frames above the 50-mm operational-error limit with an area under the receiver-operating characteristic curve of 0.71, indicating useful but incomplete self-diagnosis. Head residual-velocity jitter rose from 0.074 to 0.129 m/s. The head-error increase was 29.9 mm frontally and 18.7 mm obliquely. Controller holding added a simulated 10.4-mm wrist penalty, while dim lighting added 6.3 mm to the pooled HMD effect. In the repeatable-rig instance, adding only the HMD increased head error by 31.5 mm but lower-limb error by 0.7 mm, supporting the intended appearance-localization check.

The frozen RTMPose replication produced the same spatial ordering but a smaller simulated magnitude: head-region excess error was 22.1 mm and all-landmark excess was 7.0 mm, compared with 26.3 and 9.2 mm for MediaPipe. Because the models use different landmark definitions and score scales, estimates were reported separately rather than pooled.

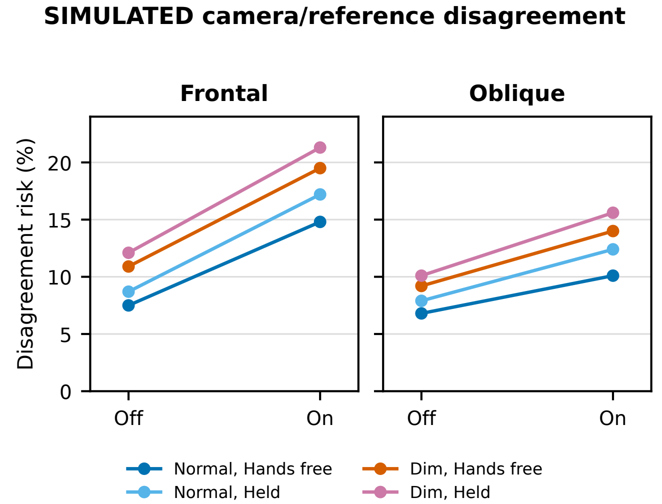

Fig. 4. Simulated camera/reference disagreement risk across view, lighting, and controller conditions. Points are the input probabilities used to test the reporting pipeline; connecting lines emphasize the paired HMD contrast.

7.3 RQ2: Derived-Quantity Agreement

The largest simulated propagation into derived kinematics occurred for shoulder ROM (Table 5). HMD-on shoulder-flexion bias was –6.9°, compared with –1.8° without the HMD; its limits of agreement also widened from 15.8° to 27.0°. Timing bias changed little and its HMD-effect interval included zero. No MDC_{95} comparison is made because an independent, movement-matched MDC has not been established.

7.4 RQ3: Decision Disagreement and Flips

The primary simulated camera/reference disagreement risk was 9.1% with the HMD off and 15.6% with it on, an adjusted risk difference of 6.5 percentage points (95% CI [4.1, 8.9]), risk ratio 1.71 (95% CI [1.41, 2.08]), $p < .001$. The HMD-associated increase was 8.4 points for the frontal view and 4.6 points for the oblique view; their interaction was -3.8 points (95% CI [-6.1, -1.5]). Controller holding amplified the HMD penalty by 2.1 points, and dim lighting raised disagreement in both HMD states (Fig. 4).

The pre-specified normal-lighting, hands-free diagnostic subset contains 1,008 HMD-off/HMD-on camera-view pairs (24 participants × 7 movements × 3 repetitions × 2 views). Reference-based equivalence criteria were satisfied by 954 pairs (94.6%). Among these, 126 (13.2%) changed camera verdict between HMD states (Table 6). Flips occurred in 27.8% of pairs within 10% of a decision boundary but in 7.4% of pairs farther away, the expected concentration if measurement error rather than a new behavioral class drives the transition.

7.5 RQ4: Mitigation at Stated Coverage

Both pre-specified mitigations reduced disagreement in the simulated HMD-on condition (Table 7). Switching from the frontal to the fixed oblique view preserved all trials and

TABLE 5
Simulated Camera–Reference Agreement for Derived Quantities

Quantity	Unit	Off bias [95% LoA]	On bias [95% LoA]	Δ [95% CI]
Shoulder-flexion ROM	deg	-1.8 [-9.7, 6.1]	-6.9 [-20.4, 6.6]	-5.1 [-7.4, -2.8]
Shoulder-abduction ROM	deg	-2.2 [-11.0, 6.6]	-7.8 [-22.6, 7.0]	-5.6 [-8.4, -2.9]
Trunk inclination	deg	0.7 [-5.9, 7.3]	3.6 [-7.8, 15.0]	2.9 [1.0, 4.8]
Trunk rotation	deg	-1.3 [-8.5, 5.9]	-3.5 [-12.9, 5.9]	-2.2 [-4.1, -0.3]
Movement duration	ms	12 [-124, 148]	24 [-164, 212]	12 [-18, 42]

TABLE 6
Simulated HMD-Off to HMD-On Decision Transitions

Off \ On	Acceptable	Error	Compensatory
Acceptable	500	44	18
Error	26	207	12
Compensatory	11	15	121

TABLE 7
Simulated HMD-On Mitigation Performance

Pipeline	Coverage %	Risk %	Δ risk [95% CI] percentage points
Frontal, no abstention	100.0	18.2	Reference
Fixed oblique	100.0	13.0	-5.2 [-7.0, -3.4]
Frontal + abstention	90.8	13.9	-4.3 [-6.2, -2.4]
Quality-selected view	94.7	11.2	-7.0 [-8.9, -5.0]

reduced disagreement by 5.2 percentage points. The pilot-locked frontal abstention rule retained 90.8% of repetitions and reduced risk by 4.3 points. Selecting the higher-quality simultaneous view and abstaining only when both views failed retained 94.7% and reduced risk by 7.0 points. These drafting values test coverage-aware reporting; they do not establish a deployment guarantee.

The secondary pose backbone also showed a reduced simulated disagreement risk: 8.4% HMD-off and 13.5% HMD-on, a 5.1-point increase (95% CI [2.9, 7.4]). Thus the direction of the demo result did not depend on selecting the backbone with the larger effect.

7.6 Runtime, Discomfort, and Safety

For validation-draft completeness, indicative deployment timings are carried over from the prior VIRTA system evaluation: pose inference required 42 ± 8 ms/frame, local WebRTC transfer 12–18 ms, rendering/animation update 58 ± 10 ms, and Azure speech synthesis 180 ± 25 ms. The sequential camera-to-visual-update subtotal is therefore approximately 112–118 ms; a spoken response is approximately 292–298 ms when speech synthesis is added. Previously reported text-to-speech naturalness was 4.3 ± 0.4 of 5 ($n = 15$). These timings are indicative carried-over measurements, not a benchmark of the exact TVCG configuration.

The approximated safety record contains no serious adverse event, no early termination, and four transient mild reports (two eye-strain episodes, one headache, and one dizziness episode), all resolving after a scheduled or requested rest. Median post-block HMD-off/HMD-on scores were 3.0/3.5 for perceived exertion, 0.5/1.5 for visual discomfort, 0.0/0.5 for nausea, and 0.5/1.0 for neck discomfort on the 0–10 scales. No score exceeded 6. These values complete the validation reporting layout and require replacement by the signed session and adverse-event logs.

8 Discussion

The simulated analysis demonstrates the argument the completed study is designed to test: a localized appearance change can be modest in aggregate position error yet consequential at the decision layer. In the drafting instance, HMD wear raised camera/reference disagreement by 6.5 percentage points, while direct HMD-off to HMD-on verdict changes reached 13.2% among reference-matched pairs. The planned mitigations also illustrate the required deployment trade-off: the fixed oblique view reduced simulated HMD-on disagreement without rejecting trials, while the two-view quality selector reduced it further at 94.7% coverage. The distinction is important. The first quantity controls for how the person actually moved in each trial; the second describes the practical instability a user or clinician would observe across visually different but kinematically matched executions.

8.1 From Occlusion to a Changed Decision

Three features of the simulated pattern form a coherent mechanism. First, the largest excess error and loss of visibility occur at the head, with smaller effects along the upper-body kinematic chain and little lower-limb change. Second, the articulated-rig pattern is similar, which is what would be expected when altered pixels rather than altered human behavior dominate. Third, shoulder ROM bias and limits of agreement worsen while task timing remains comparatively stable. A threshold using upper-body geometry is therefore more likely to change verdict than a timing rule derived from the same sequence. The same spatial ordering in the second pose backbone would make the mechanism more credible than an effect isolated to one model. Reference qualification is equally important: the central error chain is interpretable only when the inertial system's bias, drift, and response to the powered HMD remain below registered limits.

The confidence AUC of 0.71 in the simulation also illustrates why confidence cannot be treated as a correctness probability. It supports selective abstention, but a substantial subset of high-error frames would still appear confident. The final analysis should therefore report calibration and risk–coverage curves alongside confidence, not use a single universal cutoff or conceal rejected repetitions.

8.2 Implications for External Sensing

If the empirical data reproduce this pattern, three bounded design responses follow. An oblique camera around 35° may reduce frontal visor occlusion, although it does not eliminate controller or self-occlusion. Assessment systems should abstain or request a repeated movement when head/shoulder confidence and residual stability jointly cross registered limits. Finally, applications that require accurate head pose should fuse the HMD's internal head trajectory with external lower-body estimates instead of forcing the RGB model to infer an occluded face. Each response must be validated prospectively; the simulated values do not establish a deployment guarantee. In particular, the present experiment evaluates a Quest 2 and healthy adults; the geometric descriptors support replication but do not justify claims about other headsets or patient populations.

8.3 Relationship to Prior Characterization Work

Robotic evaluations of commercial HMD tracking show the value of isolating a device's sensing behavior with repeatable motion [6]. The present design transfers that logic to the camera outside the headset and then connects the error to movement-assessment decisions. It also complements generic occlusion benchmarks [3], [4]: rather than asking whether a model survives arbitrary overlap, it tests a stable, deployment-specific occluder whose location and downstream consequences are known. The error atlas and machine-readable transition data are intended to make the result reusable by future pose backbones and rehabilitation systems.

All numerical interpretations in this section are deliberately conditional on the simulated drafting instance. The completed Discussion must be regenerated from the locked empirical tables, including null or directionally different effects if those are observed. Because the decision rules are engineering operating points, even a robust empirical disagreement effect would establish measurement risk rather than rehabilitation efficacy or diagnostic validity.

9 Limitations

The study design bounds rather than eliminates several sources of uncertainty. First, one HMD model and strap configuration cannot represent every headset. The rig can establish whether a mechanism follows occlusion geometry, but generalization to other devices still requires replication across visor size, color, reflectance, and controller form. Firmware and pose-model updates can also change results; this is why versions and weights are pinned. Projected silhouette descriptors make the mechanism measurable but do not substitute for replication with geometrically different devices.

Second, inertial motion capture is a portable reference, not error-free optical ground truth. Segment calibration, magnetic disturbance, soft-tissue motion, and drift can affect joint positions and angles [37]. Fixed session registration, independent rig qualification, and sensitivity to the qualification residual distribution make these errors visible. Outcomes whose reference channel fails its registered limit are not interpreted. A future replication with a calibrated marker-based optical system would nevertheless strengthen claims about absolute 3-D error.

Third, healthy adults performing controlled repetitions do not represent the movement variability, assistive devices, fatigue, or safety constraints of a clinical population. This scope is suitable for device characterization but does not establish clinical effectiveness. Decision thresholds must therefore be described as assessment-operating points, not diagnoses, unless independently validated in the intended patient population.

Fourth, the factorial protocol cannot make HMD-on and HMD-off human movements physically identical. The same-trial reference-disagreement outcome addresses this problem for the primary decision analysis, and the direct flip analysis is restricted to reference-matched pairs, but residual behavioral differences may remain. The rig is the only arm that isolates appearance without human adaptation.

Fifth, results depend on camera placement, lighting range, clothing, body shape, and the selected pose backbones. Simultaneous frontal and oblique views test an important deployment variable, not the entire viewing sphere. The public videos, where consent permits, and frozen raw-video manifests allow future models to be evaluated on the same observations without recapturing the study.

Sixth, abstention improves retained-trial risk partly by declining difficult inputs. It is useful only when coverage remains operationally acceptable and a repeat or alternate view is available. Confidence-score scales are model-specific, so each backbone requires a separately locked rule; the risk–coverage curves are not universal calibration guarantees.

Finally, this validation version is populated with simulated or approximated cohort, pilot, qualification, geometry, safety, runtime, and outcome values. It demonstrates a complete evidentiary and reporting chain but cannot support empirical, ethical, clinical, or device-performance claims. Every such value must be reconciled against source records before submission.

10 Conclusion

HMD-on external pose estimation is a distinct measurement condition, not a minor variant of conventional no-headset evaluation. This paper specifies a reference-qualified, two-arm characterization that separates visual occlusion from behavioral change and follows landmark error through derived kinematics to assessment decisions. It also turns two practical responses—oblique viewing and selective abstention—into coverage-aware, prospectively evaluated mitigations and requires the result to replicate across two frozen pose backbones. The simulated drafting instance produces the expected diagnostic signature: large head-region degradation, smaller connected upper-body error, higher camera/reference disagreement, and lower disagreement after mitigation. These values are not empirical claims. Their purpose is to validate a reproducible route from synchronized recordings and reference qualification to the error atlas, agreement analysis, transition matrix, and risk–coverage report that will form the completed Quest 2 case study. Any rehabilitation or diagnostic claim requires separate validation in the intended population.

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